

Why do AI data centers use fresh water for cooling?

Posted by u/awesomebro5928 • 0 points • 7 comments

Why not use salt water or any type of water that we can't drink? Wouldn't all water have cooling properties? It's one of the main complaints I see with AI but it just seems like it's easily fixable.

Comments

u/Party-Ring445 • 2 points • 2025-07-06 22:00 UTC

Is the water actually used up or just passing thru for heat transfer purpose? Maybe the water was on it's way to town and just does a couple laps at the data center and comes out a couple degrees warmer as it continues its journey?

Im quite ignorant on this, someone please correct me.

u/Bobbob34 • -1 points • 2025-07-06 05:16 UTC

Salt water would destroy it in short order.

>Why not use salt water or any type of water that we can't drink? Wouldn't all water have cooling properties? It's one of the main complaints I see with AI but it just seems like it's easily fixable.

I really don't understand what happens when ppl with no knowledge, training, or experience in an area come up with some really obvious thing no one is using and think that'll work, or that something that is said to be problematic is "easily fixable" and.... what? The actual trained people are just droolingly stupid? Have never heard of the most obvious thing a child would think of?

u/awesomebro5928 • 2 points • 2025-07-06 05:39 UTC

Well that's why I'm asking. Because I don't know.

u/azuth89 • 2 points • 2025-07-06 04:55 UTC

Sure. Where does your city have the main supply of that to hook into?

Or are we expecting data centers to negotiate supply from whatever location you're imagining, build infrastructure to transport it, filter it enough to run through the cooling equipment without insta clogging it, sanitize it enough for a clean area and for human handling, desalinate it if you're talking about saltwater to avoid corrosion and conduction issues and then build the infrastructure to handle the runoff from all that filtration?

Or...they use the existing infrastructure for all that instead of entering a whole additional industry generally dominated and regulated by public bodies.

u/archpawn • 1 points • 2025-07-06 04:50 UTC

Ocean water tends to corrode stuff. And it's not like it's actually using much water. You use orders of magnitude more on food.

u/WorldTallestEngineer • 6 points • 2025-07-06 04:50 UTC

Most data centers are not next to the ocean. Salt water is extremely corrosive.

u/N4meless24- • 3 points • 2025-07-06 04:49 UTC

Absence of electrolytes, meaning less risk of mistakenly conducting electricity where it shouldn't, and consequently fry something.